

Smart Geo-Fence Alert for Fisherman

Theme:

Safety and Security for Fisherman

Introduction

A Smart IoT-based device/Application for Fishermen safety with geo-fencing, GPS tracking, and alert system.

- It ensures security by preventing border crossing and enabling real time monitoring.
- Low-cost and power efficient design suitable for fisherman with offline alert mechanism (buzzer or vibration) even without internet.

Innovation & Uniqueness

Unlike existing GPS trackers, our solution is low-cost, waterproof, solar-powered, and works offline with buzzer/vibration alerts. It integrates with a mobile app for real-time monitoring, ensuring safety even in poor network conditions

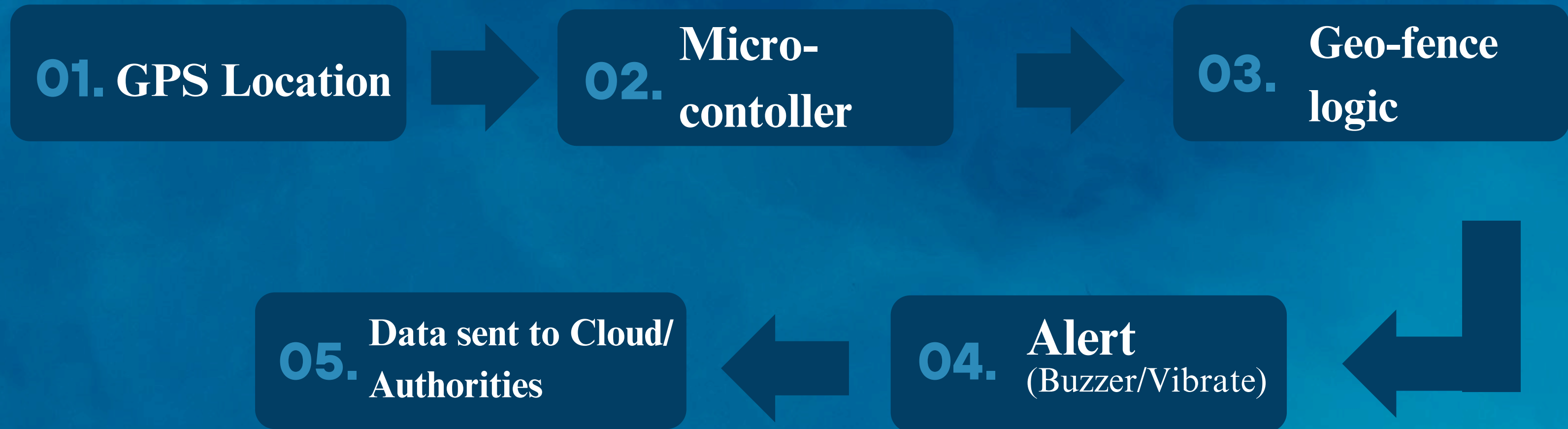


TECHNICAL APPROACH

Technologies to be used for this project;

- Hardware: GPS module, GSM module, Microcontroller(ESP32), and battery backup.
- Software: Arduino IDE (programming), MIT Application (mobile app creation) and Cloud Database (data storing).

FLOW DIAGRAM



FEASIBILITY AND VIABILITY

- Works in coastal regions with GSM network with waterproof design.
- It has challenges of signal issues in deep sea and Battery life during long fishing trips.
- Use of power-efficient hardware, Solar charging for battery and Offline alerts even without network.



IMPACT AND BENEFITS



Social Impact:

Reduces accidental border crossing and Saves lives during emergencies.



Economic Impact:

- Prevents fishermen from being arrested across borders.
- Cost-effective compared to existing solutions.



Environmental Impact:

Encourages safe and sustainable fishing without any fear of border crossing.

Research & References



- Government reports on India–Sri Lanka fishermen conflict.
- IEEE research on geo-fencing with GPS & GSM.
- Case studies from The Hindu & Indian Express on fishermen arrests.
- Studies on low-power IoT devices for remote areas.

Thank You!